

DATABASE MANAGEMENT SYSTEMS ASSIGNMENT

ONLINE SHOPPING APPLICATION

Application: Online Shopping (E-Commerce Platform)

Topic: Relational Database Design & Table Identification

1. Name and Purpose of the Application

Application Name: Online Shopping System (E-Commerce Platform)

Purpose: The Online Shopping System is a web-based application that allows customers to browse products, add them to a cart, place orders, make payments, and track deliveries. The database backing this application must reliably store information about customers, the product catalogue, shopping carts, orders, payments, shipments, and product reviews, while keeping all of this data consistent and linked together through well-defined relationships.

The goal of the database design below is to support the core workflow of an online store — a customer registers, browses products by category, adds items to a cart, checks out to create an order, pays for it, and receives a shipment — using only the tables that are actually needed to run that workflow. Every table listed has a clear, distinct role in the system.

2. List of Required Tables (Overview)

The following 11 tables together model the Online Shopping application:

- Customers – stores registered customer/user details
- Addresses – stores one or more delivery/billing addresses per customer
- Categories – organizes products into browsable groups
- Products – the product catalogue, including price and stock
- Shopping_Cart – the active cart belonging to a customer
- Cart_Items – individual products placed inside a cart
- Orders – a confirmed purchase made by a customer
- Order_Items – the line items (products) belonging to an order
- Payments – payment transaction details for an order
- Shipments – delivery/tracking details for an order
- Reviews – customer ratings and feedback on products

3. Table Definitions: Columns, Keys, and Purpose

Each table below lists its attributes, data types, key constraints (PK = Primary Key, FK = Foreign Key), and a short description of the column. A one-line purpose statement precedes each table.

3.1. Customers

Stores account and contact details for every registered customer.

Column / Attribute	Data Type	Key	Description
customer_id	INT	PK	Unique identifier for the customer
first_name	VARCHAR(50)		Customer's first name
last_name	VARCHAR(50)		Customer's last name
email	VARCHAR(100)	UNIQUE	Login email / contact email
phone	VARCHAR(15)		Contact phone number

Column / Attribute	Data Type	Key	Description
password_hash	VARCHAR(255)		Encrypted login password
registration_date	DATE		Date the account was created

3.2. Addresses

Stores one or more shipping/billing addresses linked to a customer.

Column / Attribute	Data Type	Key	Description
address_id	INT	PK	Unique identifier for the address
customer_id	INT	FK	References Customers.customer_id
address_line	VARCHAR(150)		Street / house / apartment details
city	VARCHAR(50)		City name
state	VARCHAR(50)		State or province
pincode	VARCHAR(10)		Postal / ZIP code
address_type	VARCHAR(10)		Home, Work, or Other

3.3. Categories

Groups products into browsable categories such as Electronics or Clothing.

Column / Attribute	Data Type	Key	Description
category_id	INT	PK	Unique identifier for the category
category_name	VARCHAR(50)		Name of the category, e.g. 'Footwear'
description	VARCHAR(200)		Short description of the category

3.4. Products

Holds the full catalogue of items available for purchase, including price and stock level.

Column / Attribute	Data Type	Key	Description
product_id	INT	PK	Unique identifier for the product
category_id	INT	FK	References Categories.category_id
product_name	VARCHAR(100)		Name of the product
description	VARCHAR(500)		Detailed product description
price	DECIMAL(10,2)		Selling price of the product
stock_quantity	INT		Number of units currently in stock
brand	VARCHAR(50)		Manufacturer or brand name

3.5. Shopping_Cart

Represents the active, unchecked-out cart belonging to a customer.

Column / Attribute	Data Type	Key	Description
cart_id	INT	PK	Unique identifier for the cart
customer_id	INT	FK, UNIQUE	References Customers.customer_id
created_date	DATE		Date the cart was created

3.6. Cart_Items

Stores each product placed inside a shopping cart, along with quantity.

Column / Attribute	Data Type	Key	Description
cart_item_id	INT	PK	Unique identifier for the cart item
cart_id	INT	FK	References Shopping_Cart.cart_id
product_id	INT	FK	References Products.product_id
quantity	INT		Number of units of this product in the cart

3.7. Orders

Represents a confirmed purchase made by a customer, generated at checkout.

Column / Attribute	Data Type	Key	Description
order_id	INT	PK	Unique identifier for the order
customer_id	INT	FK	References Customers.customer_id
address_id	INT	FK	References Addresses.address_id (delivery address)
order_date	DATETIME		Date and time the order was placed
order_status	VARCHAR(20)		Placed, Shipped, Delivered, or Cancelled
total_amount	DECIMAL(10,2)		Total value of the order

3.8. Order_Items

Stores each product line item that belongs to a specific order, at the price paid.

Column / Attribute	Data Type	Key	Description
order_item_id	INT	PK	Unique identifier for the order line item

Column / Attribute	Data Type	Key	Description
order_id	INT	FK	References Orders.order_id
product_id	INT	FK	References Products.product_id
quantity	INT		Number of units of this product ordered
unit_price	DECIMAL(10,2)		Price of one unit at the time of order

3.9. Payments

Records the payment transaction made against an order.

Column / Attribute	Data Type	Key	Description
payment_id	INT	PK	Unique identifier for the payment
order_id	INT	FK, UNIQUE	References Orders.order_id
payment_method	VARCHAR(20)		Card, UPI, Net Banking, or Cash on Delivery
payment_status	VARCHAR(20)		Pending, Successful, or Failed
payment_date	DATETIME		Date and time the payment was processed
amount_paid	DECIMAL(10,2)		Amount charged to the customer

3.10. Shipments

Tracks delivery details and courier status for an order.

Column / Attribute	Data Type	Key	Description
shipment_id	INT	PK	Unique identifier for the shipment
order_id	INT	FK, UNIQUE	References Orders.order_id
courier_name	VARCHAR(50)		Delivery/courier partner name
tracking_number	VARCHAR(50)		Courier tracking reference number
shipped_date	DATE		Date the order was dispatched
expected_delivery_date	DATE		Estimated date of delivery
shipment_status	VARCHAR(20)		In Transit, Delivered, or Delayed

3.11. Reviews

Stores customer ratings and written feedback for a purchased product.

Column / Attribute	Data Type	Key	Description
review_id	INT	PK	Unique identifier for the review

Column / Attribute	Data Type	Key	Description
product_id	INT	FK	References Products.product_id
customer_id	INT	FK	References Customers.customer_id
rating	INT		Star rating from 1 to 5
comment	VARCHAR(500)		Written review text
review_date	DATE		Date the review was submitted

4. Relationships Between Tables

The tables are connected through primary key / foreign key relationships that mirror how the real application behaves — one customer can have many addresses and orders, one order can contain many products, and so on. The table below summarizes each relationship.

Table A	Relationship	Table B	Description
Customers	1 : N	Addresses	A customer can save multiple delivery/billing addresses
Customers	1 : 1	Shopping_Cart	Each customer has exactly one active cart
Shopping_Cart	1 : N	Cart_Items	A cart can contain many product line items
Products	1 : N	Cart_Items	A product can appear in many customers' carts
Categories	1 : N	Products	A category groups many products
Customers	1 : N	Orders	A customer can place many orders over time
Addresses	1 : N	Orders	A saved address can be used as the delivery address for many orders
Orders	1 : N	Order_Items	An order is made up of one or more product line items
Products	1 : N	Order_Items	A product can appear in many different orders
Orders	1 : 1	Payments	Each order has exactly one payment transaction
Orders	1 : 1	Shipments	Each order has exactly one shipment record
Products	1 : N	Reviews	A product can receive many customer reviews
Customers	1 : N	Reviews	A customer can write reviews for many products

4.1 Entity Relationship Summary

Customers (1) — (N) Addresses

Customers (1) — (1) Shopping_Cart (1) — (N) Cart_Items (N) — (1) Products

Categories (1) — (N) Products

Customers (1) — (N) Orders (N) — (1) Addresses

Orders (1) — (N) Order_Items (N) — (1) Products

Orders (1) — (1) Payments

Orders (1) — (1) Shipments

Products (1) — (N) Reviews (N) — (1) Customers

5. Conclusion

The Online Shopping database design above uses eleven focused tables — Customers, Addresses, Categories, Products, Shopping_Cart, Cart_Items, Orders, Order_Items, Payments, Shipments, and Reviews — to represent the complete lifecycle of an e-commerce transaction: registration, browsing, cart management, order placement, payment, delivery tracking, and post-purchase feedback.

Splitting Cart_Items from Shopping_Cart and Order_Items from Orders avoids repeating group data (multiple products per cart or order) inside a single row, which keeps the design in line with normalization principles. Each table has a single, clear responsibility, and every relationship reflects a real business rule of how an online store operates — demonstrating how a real-world application can be modelled as a relational database.